CrewAI-Gemini.ipynb - Colab
Code error fix request

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CrewAI-Gemini.ipynb

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```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
!pip install --q crewai
!pip install --q -U duckduckgo-search
!pip install --q langchain_google_genai
!pip install --q langchain-community
!pip install --q -U ddgs
```

```
    ──────────────────────────────────────── 43.6/43.6 kB 3.7 MB/s eta 0:00:00
    ──────────────────────────────────────── 67.3/67.3 kB 6.6 MB/s eta 0:00:00
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
    ──────────────────────────────────────── 67.7/67.7 kB 5.9 MB/s eta 0:00:00
  ──────────────────────────────────────── 642.9/642.9 kB 24.9 MB/s eta 0:00:00
  ──────────────────────────────────────── 19.9/19.9 MB 90.6 MB/s eta 0:00:00
  ──────────────────────────────────────── 160.9/160.9 kB 15.3 MB/s eta 0:00:00
  ──────────────────────────────────────── 60.0/60.0 kB 6.2 MB/s eta 0:00:00
  ──────────────────────────────────────── 5.6/5.6 MB 136.9 MB/s eta 0:00:00
  ──────────────────────────────────────── 250.1/250.1 kB 24.9 MB/s eta 0:00:00
  ──────────────────────────────────────── 21.7/21.7 MB 125.2 MB/s eta 0:00:00
  ──────────────────────────────────────── 278.2/278.2 kB 22.2 MB/s eta 0:00:00
  ──────────────────────────────────────── 45.5/45.5 kB 4.5 MB/s eta 0:00:00
  ──────────────────────────────────────── 358.8/358.8 kB 34.9 MB/s eta 0:00:00
```

Variables Terminal 13:46 T4 (Python 3)

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RAM Disk

```
google-colab 1.0.0 requires requests==2.32.4, but you have requests 2.32.5 which is incompatible.
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 41.6/41.6 kB 3.2 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 161.7/161.7 kB 11.1 MB/s eta 0:00:00
```

[ ]

```python
import os
from langchain_google_genai import ChatGoogleGenerativeAI
from crewai import Agent, Task, Crew, Process

# Define llm globally here to avoid scope issues and allow direct ChatGoogleGenerativeAI instance usage
llm = ChatGoogleGenerativeAI(model="gemini-1.5-flash",
                             verbose = True,
                             temperature = 0.2,
                             google_api_key="AIzaSyD4wjxz1ZuoTatsxxXSPFM9jxmNoda4A2A")
```

[ ]

```python
# This cell previously defined 'llm' as a dictionary, which was conflicting with the 'ChatGoogleGenerativeAI' instance.
# It is now commented out to ensure the correct 'llm' from Y05pqQmpOsCv is used.
```

[ ]

```python
# This cell previously inspected the 'llm' dictionary, which is no longer needed after the fix.
```

[ ]

```python
from langchain_community.tools.ddg_search import DuckDuckGoSearchRun

# DuckDuckGoSearchRun already inherits from langchain_core.tools.BaseTool
search_tool = DuckDuckGoSearchRun()
```

[ ]

```python
!pip install git+https://github.com/huggingface/smolagents.git
```

{} Variables | Terminal | ✓ 13:46 | T4 (Python 3)

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RAM
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```
  Building wheel for smolagents (pyproject.toml) ... done
  Created wheel for smolagents: filename=smolagents-1.24.0.dev0-py3-none-any.whl size=148615 sha256=25d14089ef5f53ac67b35057d040efa9cc18ec4cde51da669c59deec31978be4
  Stored in directory: /tmp/pip-ephem-wheel-cache-qdzh6q6t/wheels/9a/fd/12/d7f10c4454bc4b2978bcc7e8554b2bfa4cbd13e46a13249270
Successfully built smolagents
Installing collected packages: smolagents
Successfully installed smolagents-1.24.0.dev0
```

[ ]

```
!pip install litellm
```

```
Collecting litellm
  Downloading litellm-1.80.7-py3-none-any.whl.metadata (30 kB)
Requirement already satisfied: aiohttp>=3.10 in /usr/local/lib/python3.12/dist-packages (from litellm) (3.13.2)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from litellm) (8.3.1)
Collecting fastuuid>=0.13.0 (from litellm)
  Downloading fastuuid-0.14.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (1.1 kB)
Collecting grpcio<1.68.0,>=1.62.3 (from litellm)
  Downloading grpcio-1.67.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (3.9 kB)
Requirement already satisfied: httpx>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from litellm) (0.28.1)
Requirement already satisfied: importlib-metadata>=6.8.0 in /usr/local/lib/python3.12/dist-packages (from litellm) (8.7.0)
Requirement already satisfied: jinja2<4.0.0,>=3.1.2 in /usr/local/lib/python3.12/dist-packages (from litellm) (3.1.6)
Requirement already satisfied: jsonschema<5.0.0,>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from litellm) (4.25.1)
Requirement already satisfied: openai>=2.8.0 in /usr/local/lib/python3.12/dist-packages (from litellm) (2.8.1)
Requirement already satisfied: pydantic<3.0.0,>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from litellm) (2.12.3)
Requirement already satisfied: python-dotenv>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from litellm) (1.2.1)
Requirement already satisfied: tiktoken>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from litellm) (0.12.0)
Requirement already satisfied: tokenizers in /usr/local/lib/python3.12/dist-packages (from litellm) (0.22.1)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp>=3.10->litellm) (2.6.1)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp>=3.10->litellm) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp>=3.10->litellm) (25.4.0)
```

Variables Terminal 13:46 T4 (Python 3)

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Code error fix request

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RESTART SESSION

[4] 36s

```
!pip install -q crewai crewai-tools
```

```
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 43.6/43.6 kB 3.5 MB/s eta 0:00:00
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  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
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   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 765.0/765.0 kB 55.4 MB/s eta 0:00:00
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   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 147.8/147.8 kB 12.2 MB/s eta 0:00:00
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   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 60.0/60.0 kB 5.0 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 5.6/5.6 MB 94.3 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 24.1/24.1 MB 115.0 MB/s eta 0:00:00
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   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 57.6/57.6 kB 5.1 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 64.7/64.7 kB 5.1 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 250.1/250.1 kB 20.8 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 21.7/21.7 MB 34.7 MB/s eta 0:00:00
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   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 278.2/278.2 kB 22.1 MB/s eta 0:00:00
```

Variables Terminal 13:46 T4 (Python 3)

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Reasoning: Now that `litellm` is installed, I will re-execute the agent definition code (from cell `63cb3af8`) to confirm that the `ImportError` related to `LiteLLM` is resolved and the `researcher` agent can be successfully initialized without the `tools` parameter.

[7] 8s

```
!pip install -q crewai crewai-tools langchain-google-genai langchain-community ddgs duckduckgo_search
```

[12] 0s

```
from langchain_google_genai import ChatGoogleGenerativeAI
import os, getpass

# Set API key safely
if "GOOGLE_API_KEY" not in os.environ:
    os.environ["GOOGLE_API_KEY"] = getpass.getpass("Enter your Google API key: ")

# Create LLM instance
llm = ChatGoogleGenerativeAI(
    model="gemini-1.5-flash",
    temperature=0.2,
    verbose=True,
    google_api_key=os.environ["GOOGLE_API_KEY"]
)

print("LLM created successfully!")
```

```
LLM created successfully!
```

Variables Terminal   13:46 T4 (Python 3)

CrewAI-Gemini.ipynb

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LLM created successfully!

[24] 8s

```
# Single combined Colab cell — use models/gemini-2.5-flash (copy & run)
import os, getpass, traceback, inspect, asyncio
import nest_asyncio
nest_asyncio.apply()

# 0) Ensure GOOGLE_API_KEY is set (hidden input if needed)
if not os.environ.get("GOOGLE_API_KEY"):
    key = getpass.getpass("Enter your Google API key (hidden): ")
    if not key:
        raise SystemExit("No key provided. Aborting.")
    os.environ["GOOGLE_API_KEY"] = key

MODEL_ID = "models/gemini-2.5-flash"  # use the exact model id from your models list

# 1) Create LangChain Gemini LLM
try:
    from langchain_google_genai import ChatGoogleGenerativeAI
    llm = ChatGoogleGenerativeAI(
        model=MODEL_ID,
        temperature=0.2,
        verbose=True,
        google_api_key=os.environ.get("GOOGLE_API_KEY")
    )
    print("✅ Created ChatGoogleGenerativeAI with", MODEL_ID)
except Exception:
    print("❌ Failed to create ChatGoogleGenerativeAI. Traceback:")
```

Variables Terminal 13:46 T4 (Python 3)

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```
✅ Created ChatGoogleGenerativeAI with models/gemini-2.5-flash
✅ CrewAI wrapper created.
✅ Agents created (researcher, writer).
──────────────── LiteAgent Started ────────────────

 LiteAgent Session Started
 Name: Generative AI & llm model developer
 id: f51b0066-af6d-4352-9b46-83ab139bd5ae
 role: Generative AI & llm model developer
 goal: How a Non Technical can improve an llm model
 backstory: Summarize detailed roadmap to become llm developer and how to land a job.
 tools: []
 verbose: True
 Tool Args:

✅ Used method: kickoff

--- AGENT OUTPUT ---

Here are 6 bullet points summarizing how a non-technical person can improve skills to work with LLMs and land a job:

*   **Master Prompt Engineering:** Dedicate time to becoming an expert in crafting effective prompts. This involves understanding different prompting techniques (e.g., few-s
*   **Leverage Existing Domain Expertise:** Identify how your current non-technical knowledge (e.g., marketing, law, healthcare, education, customer service, content creatic
*   **Focus on AI Product & User Experience (UX):** Develop a strong understanding of how LLMs can solve real-world problems for users. This involves thinking about user nee
*   **Cultivate AI Ethics & Responsible AI Understanding:** Educate yourself on the ethical implications, potential biases, safety concerns, and responsible deployment princ
*   **Build a Practical Portfolio of LLM Applications:** Create a portfolio demonstrating your ability to use LLMs to solve problems, even without coding. This could include
*   **Network and Communicate Your Value:** Actively participate in AI communities, attend webinars, and connect with professionals in the field. Practice articulating how y
──────────────── ✅ Agent Final Answer ────────────────
```

Variables | Terminal | ✓ 13:46 | T4 (Python 3)

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```
✅ Agent Final Answer

Agent: Generative AI & llm model developer

Final Answer:
Here are 6 bullet points summarizing how a non-technical person can improve skills to work with LLMs and land
a job:

*   **Master Prompt Engineering:** Dedicate time to becoming an expert in crafting effective prompts. This
involves understanding different prompting techniques (e.g., few-shot, chain-of-thought, persona-based),
testing outputs, iterating on prompts to refine responses, and learning how to elicit specific behaviors from
LLMs. This is a highly valuable non-technical skill directly impacting LLM utility.
*   **Leverage Existing Domain Expertise:** Identify how your current non-technical knowledge (e.g.,
marketing, law, healthcare, education, customer service, content creation) can be applied to improve LLM
outputs or design LLM-powered solutions within that specific field. Your subject matter expertise is crucial
for guiding LLM application, evaluating accuracy, and ensuring relevance.
*   **Focus on AI Product & User Experience (UX):** Develop a strong understanding of how LLMs can solve
real-world problems for users. This involves thinking about user needs, designing intuitive interfaces for
LLM interaction, evaluating the quality and usability of LLM outputs from a user perspective, and
contributing to product strategy for AI-powered tools.
*   **Cultivate AI Ethics & Responsible AI Understanding:** Educate yourself on the ethical implications,
potential biases, safety concerns, and responsible deployment principles of LLMs. Roles in AI governance,
policy, content moderation, and risk assessment are increasingly important and often require strong
non-technical judgment and critical thinking.
*   **Build a Practical Portfolio of LLM Applications:** Create a portfolio demonstrating your ability to use
LLMs to solve problems, even without coding. This could include case studies of prompt engineering challenges
you've tackled, analyses of LLM outputs for specific tasks, proposals for LLM-powered solutions in your
domain, or examples of content you've refined using LLMs. Document your process and results clearly.
*   **Network and Communicate Your Value:** Actively participate in AI communities, attend webinars, and
connect with professionals in the field. Practice articulating how your non-technical skills, domain
expertise, and understanding of LLM capabilities can contribute to projects, bridging the gap between
technical development and real-world business application.
```

Variables | Terminal | ✓ 13:46 | T4 (Python 3)